

Comparison of Models for Treating Angular Momentum in RRKM Calculations with Vibrator Transition States. Pressure and Temperature Dependence of Cl + C₂H₂ Association

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The RRKM expression for the microcanonical unimolecular rate constant depends on how the K rotational quantum number is treated. The limiting cases are to treat K as either adiabatic or active. The former occurs when K is a good quantum number for describing the vibrational/rotational energy levels. The latter case arises when coriolis coupling extensively mixes the $(2J + 1) K$ levels. It is not necessary that K be simultaneously active or adiabatic for both the transition state and reactant molecule. Thus, four models which explicitly treat K with its proper limits, i.e. $-J \leq K \leq J$, can be formulated for the RRKM unimolecular rate constant. It is found that the calculated pressure- and temperature-dependent rate constant for Cl + C₂H₂ → C₂H₂Cl association is highly sensitive to the treatment of the K quantum number in RRKM theory. For example, to fit the association rate constant at low pressure using the model with K adiabatic in the transition state and active for the molecule requires a value for the collision efficiency β_c twice the size needed when K is treated as active in both the molecule and transition state. These results indicate that until the proper model for treating K is identified it will be difficult to determine unambiguous values for β_c by fitting experimental data. To assist in interpreting the experimental kinetics for the Cl + C₂H₂ → C₂H₂Cl system, ab initio calculations were performed to determine the structure, vibrational frequencies, and energy for C₂H₂Cl. The ab initio G2 value for the 0 K heat for Cl + C₂H₂ association is -16.855 kcal/mol, which agrees with semiempirical estimates of the heat of reaction. To fit the low-pressure experimental rate constants for Cl + C₂H₂ association in air requires values for β_c which range from 0.4 to 1.0. However, none of the RRKM models for treating K provides an adequate fit to all the temperature- and pressure-dependent experimental rate constants. Possible explanations for this finding include an inadequate treatment of anharmonicity, non-RRKM rate constants for C₂H₂Cl dissociation, and imperfect extrapolations of the experimental rate constants to the high- and low-pressure asymptotic limits.

I. Introduction

Studies of gas-phase reactions of Cl atoms with organic species are important, since such reactions are significant contributors to the loss of Cl atoms and non-methane hydrocarbons in the atmosphere.^{1–3} In particular, rate constants are needed for these reactions to expand a broad kinetic database, which can be used to model chemical processes in the atmosphere and in laboratory reactors designed to simulate the atmosphere.^{4–7}

There have been a limited number of measurements of the pressure and temperature dependence of the rate constant for the reaction of Cl atoms with acetylene^{8–14}



In two very recent studies^{13,14} a relative rate technique was used to measure the rate constant for this reaction over the 10–3000-Torr pressure range and for temperatures between 230 and 370 K. The rate of reaction 1 was measured relative to that for the pressure-independent abstraction reaction Cl + C₂H₆ → C₂H₅ + HCl, which has a well-established temperature-dependent rate constant. Assuming a temperature-independent center broadening factor of 0.6 in the Troe formalism,¹⁵ the measured rate constants were used to determine the following temperature-dependent high- and low-pressure limiting rate constants for reaction 1 in air:¹⁴

$$k_\infty(T) = 2.13 \times 10^{-10} (T/300)^{-1.045} \text{ cm}^3/(\text{molecule s}) \quad (2a)$$

$$k_0(T) = 5.4 \times 10^{-30} (T/300)^{-2.09} \text{ cm}^6/(\text{molecule}^2 \text{ s}) \quad (2b)$$

Both the high- and low-pressure limiting rate constants have a negative temperature dependence. At the most completely studied temperature, 295 K, the limiting rate constants, with statistical uncertainties, are $k_\infty(295) = (1.9 \pm 0.3) \times 10^{-10} \text{ cm}^3/(\text{molecule s})$ and $k_0(295) = (6.1 \pm 0.8) \times 10^{-30} \text{ cm}^6/(\text{molecule}^2 \text{ s})$.

In this study we have used the Rice–Ramsperger–Kassel–Marcus (RRKM) formalism¹⁶ to analyze the pressure and temperature dependence of reaction 1. Such an investigation is important for several reasons. First, the results of the RRKM analysis can be compared with that made previously with the Troe formalism.^{14,15} Second, questions concerning the applicability of RRKM theory to a Cl-atom plus alkyne reactive system like reaction 1 have arisen^{17–20} from experimental dynamical studies which indicate that energy distribution only occurs between a few degrees of freedom in the Cl-alkyne complex. Lee and co-workers^{18,19} made this conclusion from crossed molecular beam studies of substitution reactions like Br + C₂H₂Cl₂ → C₂H₂ClBr + Cl and F + C₂H₄ → C₂H₃F + H. The same conclusion was reached by Bersohn and co-workers²⁰ from measurements of the product translational energy for the exchange reaction H + RD → RH + D, where RD is deuterated acetylene, methylacetylene, ethylene, and propylene. Given these questions concerning the validity of RRKM theory, a direct comparison between RRKM theory and experiment for reaction 1 is of considerable interest.

The remainder of this paper is organized in the following manner. In section II the RRKM vibrator transition-state model is outlined. Different treatments of rotational angular momentum are discussed and expressions for calculating the association rate constant versus temperature and pressure are also given in this

section. The potential energy surface, reaction energetics, and transition-state structure for the $\text{Cl} + \text{C}_2\text{H}_2 \rightleftharpoons \text{C}_2\text{H}_2\text{Cl}$ system are described in section III. Theoretical calculations of the temperature and pressure dependence of the $\text{Cl} + \text{C}_2\text{H}_2$ association rate constant are compared with experimental results in section IV. The manuscript ends with section V, a discussion.

II. RRKM Vibrator Transition State Model and the Association Rate Constant

For the $\text{C}_2\text{H}_2\text{Cl} \rightarrow \text{C}_2\text{H}_2 + \text{Cl}$ unimolecular reaction bending vibrational modes of the Cl atom have low frequencies in the $\text{C}_2\text{H}_2 \cdots \text{Cl}$ transition state. Such modes are often called "transitional modes",^{21,22} and two approaches have been used for treating them in RRKM calculations of unimolecular dissociation rate constants. In one approach, called the flexible transition-state model,^{23,24} the long-range Hamiltonian for the intermolecular interaction between the dissociation products is used to treat the transitional modes as classical hindered rotors with proper angular momentum constraints. In the second approach, the vibrator transition-state model,^{24,25} the transitional modes are represented as quantum harmonic oscillators, as are other vibrational degrees of freedom. This latter method is used here. For CH_4 dissociation, which has a potential energy surface with general attributes like those for $\text{C}_2\text{H}_2\text{Cl}$ dissociation, the flexible and vibrator transition-state models give rate constants which agree to within 30%.²⁴

A. Treatment of Rotational Angular Momentum. In the RRKM theory of unimolecular reactions the energy- and angular-momentum-dependent unimolecular rate constant is determined from the sum of states at the transition state and the density of states for the reactant molecule. The most accurate calculation of this sum and density involves determining for the transition state and molecule vibrational-rotational levels with all anharmonic and coriolis coupling terms included. For most unimolecular reactions this is a formidable if not impossible task.²⁶ Thus, one resorts to the use of models for calculating the sum of states for the transition state and the density of states for the molecule.

A widely used RRKM model for determining the sum and density is to assume separable vibrational and rotational degrees of freedom, treated as harmonic oscillators and rigid rotors, respectively.²⁷ The rotational energy levels for an "almost symmetric top" rigid rotor are given by²⁸

$$E_r(J,K) = (I_a^{-1} + I_b^{-1})[J(J+1) - K^2]\hbar^2/4 + K^2\hbar^2/2I_c \quad (3)$$

where $I_a \approx I_b \neq I_c$. The quantum number J is for the total angular momentum, and K represents the projection of J onto the symmetry axis. Their values are $J = 0, 1, \dots, \infty$ and $K = -J, -(J-1), \dots, 0, \dots, J-1, J$. A molecule containing total vibrational-rotational energy E , in a particular J, K level, has a vibrational density of states denoted by $N[E - E_r(J,K)]$. Similarly, the transition states sum of states for the same E, J , and K is denoted by $G^*[E - E_0 - E_r^*(J,K)]$, where E_0 is the unimolecular threshold. For this model K is considered *adiabatic* for both the *transition state* and *molecule*, and the model is identified by *adiabatic/adiabatic*. Its RRKM rate constant is given by^{24,25}

$$k(E,J,K) = G^*[E - E_0 - E_r^*(J,K)]/hN[E - E_r(J,K)] \quad (4)$$

The quantum number J represents the total angular momentum and is a constant of the motion. However, coriolis coupling can mix the $2J + 1$ K levels for a particular value of J and destroy K as a good quantum number.²⁹⁻³⁴ As a result, the above K adiabatic model would be inappropriate for calculating the rate constant. Maintaining the rigid rotor model, but allowing mixing of the $2J + 1$ K levels results in $2J + 1$ mixed levels each with the same molecular density of states, denoted by

$$N_K(E,J) = \sum_{K=-J}^J N[E - E_r(J,K)]/(2J + 1) \quad (5a)$$

$$G_K^*(E,J) = \sum_{K=-J}^J G^*[E - E_0 - E_r^*(J,K)]/(2J + 1) \quad (5b)$$

In these K -mixed models, K is considered an *active* degree of freedom.

In previous applications of RRKM theory a degree of freedom, treated as adiabatic or active in the molecule, is treated the same way in the transition state.²⁷ Thus, if K is active for the molecule, it would also be active for the transition state.³⁵ However, there is no need to require such an assumption. At low levels of excitation, as for thermal experiments, the energy content of the transition state is much lower than that of the molecule, so that the extent of coriolis coupling differs between the molecule and transition state. One likely possibility is K active for the molecule but adiabatic for the transition state. If there are significant frequency and structural changes in going from the molecule to transition state, K could be adiabatic for the molecule but active for the transition state. Thus, using terminology from adiabatic rate theory,^{29,36} the degree of mixing between the K levels may change in moving along the reaction path from the potential minimum to the transition state. Therefore, it seems particularly worthwhile to describe *mixed mode RRKM models* in which K is treated differently in the molecule and transition state.

If K is *active* for both the *transition state* and *molecule* (an *active/active* model), the RRKM rate constant only depends on E and J and is given by^{24,25}

$$k(E,J) = G_K^*(E,J)/hN_K(E,J) = \sum_{K=-J}^J G^*[E - E_0 - E_r^*(J,K)]/h \sum_{K=-J}^J N[E - E_r(J,K)] \quad (6)$$

In eqs 4 and 6 K is either completely adiabatic or active. Two additional models can be formulated for the unimolecular rate constant by treating K differently in the reactant molecule and transition state. If K is *adiabatic* for the *transition state* and *active* for the *molecule* (an *adiabatic/active* model), the unimolecular rate constant is

$$k(E,J,K) = G^*[E - E_0 - E_r^*(J,K)]/hN_K(E,J) \quad (7)$$

where $N_K(E,J)$ is given by eq 5a. In contrast to this model, K may be *active* for the *transition state* and *adiabatic* for the *molecule* (an *active/adiabatic* model) so that

$$k(E,J,K) = G_K^*(E,J)/hN[E - E_r(J,K)] \quad (8)$$

For a microcanonical ensemble, with fixed E and J , the RRKM unimolecular rate constant is given by $k(E,J)$ in eq 6.²⁵ It is straightforward to show that the $k(E,J,K)$ expressions in eqs 4, 7, and 8 yield this $k(E,J)$. The microcanonical rate constant with K treated adiabatically is written as²⁵

$$k(E,J) = \sum_{K=-J}^J P(K) k(E,J,K) \quad (9)$$

where $P(K)$ is the appropriate probability. If K is adiabatic for the molecule, $P(K)$ is given by

$$P(K) = N[E - E_r(J,K)] / \sum_{K=-J}^J N[E - E_r(J,K)] \quad (10)$$

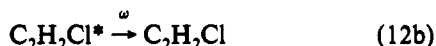
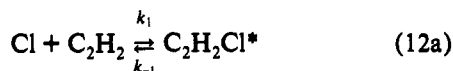
Inserting this $P(K)$ into eq 9 and combining it with the $k(E,J,K)$ in either eq 4 or 8 gives the $k(E,J)$ expression in eq 6. For the

$k(E, J, K)$ rate constant with K active for the molecule, eq 7

$$P(K) = N_K(E, J) / \sum_{K=-J}^J N[E - E_r(J, K)] = 1/(2J + 1) \quad (11)$$

Inserting this $P(K)$ and eq 7 into eq 9 gives eq 6.

B. Association Rate Constant. The mechanism for the reaction of chlorine atoms with acetylene can be written as



Based on this mechanism, RRKM theory can be used to calculate the Cl + C₂H₂ association rate constant versus temperature and pressure. The different models described above give different expressions for the association rate constant. For the active/active model, eq 6, the association rate constant can be written as

$$k_a(T, P) = \frac{\sigma\omega}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{\infty} \frac{(2J+1)G^*(E, J)}{\omega + k_{-1}(E, J)} e^{-E/k_B T} \quad (13)$$

where σ is the reaction path degeneracy, $Q_{\text{C}_2\text{H}_2}$ is the partition function for acetylene's internal degrees of freedom, Q_{rel} is the Cl + C₂H₂ relative translational energy partition function, $G^*(E, J) = \sum_{K=-J}^J G_K^*(E, J)$ is the total sum of states for the transition state at E and J (see eq 5b), $k_{-1}(E, J)$ is given by eq 6, and ω is the collision frequency. For the three models with K treated as adiabatic, eq 13 does not represent $k_a(T, P)$. The adiabatic/adiabatic model (i.e., eq 4) has

$$k_a(T, P) = \frac{\sigma\omega}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{\infty} (2J+1) \sum_{K=-J}^J \frac{G^*(E, J, K) e^{-E/k_B T}}{\omega + G^*(E, J, K)/hN(E, J, K)} \quad (14)$$

where

$$G^*(E, J, K) = G^*[E - E_0 - E_r^*(J, K)]$$

$$N(E, J, K) = N[E - E_r(J, K)]$$

For the adiabatic/active model (i.e., eq 7) the association rate constant is

$$k_a(T, P) = \frac{\sigma\omega}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{\infty} (2J+1) \sum_{K=-J}^J \frac{G^*(E, J, K) e^{-E/k_B T}}{\omega + G^*(E, J, K)/hN_K(E, J)} \quad (15)$$

Finally, for the active/adiabatic model (i.e., eq 8), one finds that

$$k_a(T, P) = \frac{\sigma\omega}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{\infty} (2J+1) \sum_{K=-J}^J \frac{G_K^*(E, J) e^{-E/k_B T}}{\omega + G_K^*(E, J)/hN(E, J, K)} \quad (16)$$

It is relatively straightforward to show that, in the $\omega \rightarrow \infty$ high-pressure limit, eqs 13–16 give the same unimolecular rate constant, if the variational criterion is unimportant in choosing the transition state so that the position of the transition state does not vary with energy and angular momentum. The resulting

expression for the high-pressure association rate constant is

$$k_{\infty}(T) = \frac{\sigma}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{\infty} (2J+1) \sum_{K=-J}^J G^*(E, J, K) e^{-E/k_B T} \quad (17)$$

Since the Laplace transform $\sum (2J+1) \sum G^*(E, J, K)$ in eq 17 is the partition function Q^* for the transition state multiplied by $k_B T$,^{27b} eq 17 can be written as

$$k_{\infty}(T) = \frac{\sigma k_B T}{h} \frac{Q^*}{Q_{\text{C}_2\text{H}_2}Q_{\text{rel}}} e^{-E_0/k_B T} \quad (18)$$

where E_0 is the energy difference between the transition state and reactant zero point energy levels. If the variational criterion is important in choosing the transition-state structure, $k_{\infty}(T)$ still has the same mathematical form as given by eq 17, but the term $\sum G^*(E, J, K)$ depends on whether K is adiabatic or active in the transition state.²⁵ For K adiabatic, a different transition state is found at each E, J , and K by finding the minimum in $G^*(E, J, K)$ along the reaction path. However, for K active, the sum $G^*(E, J) = \sum G^*(E, J, K)$ is minimized. The K adiabatic model will give a $k_{\infty}(T)$ less than or equal to that of the K active model.²⁵ If the variational criterion is important, neither the K adiabatic nor active model gives a $k_{\infty}(T)$ that can be expressed as eq 18.

In the $\omega \rightarrow 0$ low-pressure limit, $k_a(T, P)$ is insensitive to the variational criterion but does depend on whether the K quantum number is treated as adiabatic or active. For the models with K active for the transition state, which give the $k_a(T, P)$ in eqs 13 and 16, the low-pressure rate constant is written as

$$k_0(T, P) = \frac{\sigma\omega}{Q_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{J_{\text{max}}(E)} (2J+1) \sum_{K=-J}^J N(E, J, K) e^{-E/k_B T} \quad (19)$$

where $J_{\text{max}}(E)$ is chosen according to the constraint $G^*(E, J) > 0$. The origin of this constraint is that, to form the energized molecule with either density $N(E, J)$ in eq 13 or density $N(E, J, K)$ in eq 16, the transition-state sum of states $G^*(E, J) = \sum G^*(E, J, K)$ must be greater than zero.

The association rate constant expressions with K adiabatic in the transition state, eqs 14 and 15, give low-pressure rate constant expressions which differ and are also not that same as eq 19 for K active in the transition state. First consider $k_a(T, P)$ in eq 14, where K is adiabatic for both the transition state and molecule. The low-pressure rate constant is still expressed in the same form as eq 19; i.e.

$$k_0(T, P) = \frac{\sigma\omega}{Q_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{J_{\text{max}}(E)} (2J+1) \sum_{K(E, J)} N(E, J, K) e^{-E/k_B T} \quad (20)$$

where the summation over $K(E, J)$ only includes values of K for which $G^*(E, J, K) > 0$. The range of K values included in the summation depends on whether the transition state is a prolate or oblate top. For a prolate top $K(E, J)$ ranges between $-K_{\text{max}}(E, J)$ and $K_{\text{max}}(E, J)$, where $K_{\text{max}} \leq J$. However, for an oblate top $|K(E, J)|$ ranges between J and $K_{\text{min}}(E, J)$, where $K_{\text{min}} \geq 0$. For the association rate constant expression in eq 15, where K is adiabatic for the transition state and active for the molecule, the low-pressure limiting rate constant is

$$k_0(T, P) = \frac{\sigma\omega}{Q_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{J_{\text{max}}(E)} (2J+1) \sum_{K(E, J)} N_K(E, J) e^{-E/k_B T} \quad (21)$$

where $N_K(E, J)$ is given by eq 5a and $K(E, J)$ is treated the same as in eq 20. However, in eq 20 the sum is over the $N(E, J, K)$,

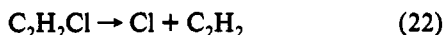
while in eq 21 the sum is over $N_K(E, J)$ which is an average of the $N(E, J, K)$. Eqs 19–21 can be summarized in the following manner. At low pressures the association rate constant is proportional to the number of $C_2H_2Cl^*$ states that can be formed by $Cl + C_2H_2$ association. With K adiabatic in the transition state this number is smaller than that with K active in the transition state.

For the calculations reported here, eqs 13–21 were solved numerically by eight-point Gauss-Laguerre integration.³⁷ Several of the numerical integrations were also performed with the trapezoid method using step sizes of 50 and 100 cm^{-1} . These two step sizes gave the same result to within three significant figures,³⁸ which was the same as that obtained from the Gauss-Laguerre integration. The microcanonical rate constants $k(E, J)$ and $k(E, J, K)$ were calculated by using the Beyer-Swinehart algorithm³⁹ to determine the sum of states and the Whitten-Rabinovitch approximation^{27a} to find the density of states.

C. Collision Stabilization Frequency. The collision frequency ω for stabilizing $C_2H_2Cl^*$ can be written as $\omega = \beta_c Z_{LJ} P_M$,⁴⁰ where β_c is the collision efficiency, Z_{LJ} is the Lennard-Jones collision frequency, and P_M is the pressure of the bath gas air which deactivates $C_2H_2Cl^*$. In calculating Z_{LJ} an effective collision cross section is used, which equals the hard-sphere cross section times the reduced collision integral $\Omega_{AM}^{(2,2)}$.⁴¹ The hard-sphere collision diameter σ and van der Waals potential energy minimum ϵ/k for $C_2H_2Cl^*$ were estimated from the σ and ϵ/k values for C_2H_4 , CH_4 , and CH_3Cl ;⁴² i.e. for $C_2H_2Cl^*$ $\sigma = 4.61$ Å and $\epsilon/k = 415$ K. Values of σ and ϵ/k for the components of air⁴³ are used to give $\sigma = 3.62$ Å and $\epsilon/k = 97.0$ K for air. The effective $C_2H_2Cl^*$ -air collision diameter varies from 4.91 Å at 252 K to 4.53 Å at 370 K.

III. Reaction Energetics and Potential Energy Surface

A. Reaction Energetics. There have been no previous studies of the energetics for the reaction



Thus, the 0 K heat of formation for this reaction ΔH°_0 was estimated by two different approximate relationships and calculated by an ab initio quantum chemical theory. In one of the approximate approaches the difference between the ΔH°_0 values for $C_2H_4Cl \rightarrow Cl + C_2H_4$ and $C_2H_2Cl \rightarrow Cl + C_2H_2$ was assumed to be the same as that between $C_2H_5 \rightarrow H + C_2H_4$ and $C_2H_3 \rightarrow C_2H_2 + H$. Using the experimental ΔH°_0 values of 21.2,⁴⁴ 38.1,⁴⁵ and 33.1 kcal/mol^{45–48} for $C_2H_4Cl \rightarrow Cl + C_2H_4$, $C_2H_5 \rightarrow H + C_2H_4$, and $C_2H_3 \rightarrow C_2H_2 + H$, respectively, gives a value of $\Delta H^\circ_0 = 16.2$ kcal/mol for $C_2H_2Cl \rightarrow Cl + C_2H_2$. In the second approximate approach the assumption was made that the $H-C_2H_3$ ⁴⁶ and $H-C_2H_2Cl$ bond dissociation energies are the same. Since the heats of formation of H and C_2H_3Cl are known,⁴⁵ this assumption yields an estimate of the C_2H_2Cl heat of formation which, when combined with the Cl and C_2H_2 heats of formation,⁴⁵ gives $\Delta H^\circ_0 = 14.5$ kcal/mol for $C_2H_2Cl \rightarrow Cl + C_2H_2$.

The C_2H_2Cl radical has both a cis equilibrium configuration in which the two H atoms are on the same side of the $C=C$ double bond and a trans equilibrium configuration. Ab initio calculations were performed at the HF/6-31G* and MP2/6-31G* levels of theory to determine the structures of cis- and trans- C_2H_2Cl . The calculated structures are depicted in Figure 1. Harmonic vibrational frequencies for the cis and trans configurations were calculated at the HF/6-31G* level and are listed in Table I along with anharmonic frequencies for the $n = 0 \rightarrow 1$ transition, which are estimated by multiplying the harmonic frequencies by a 0.8929 scale factor.⁴⁹ For completeness ab initio calculations were also performed for C_2H_2 and the results are included in Table I. The moments of inertia calculated from the MP2/6-31G* structures for C_2H_2Cl and C_2H_2 are also listed in this table.

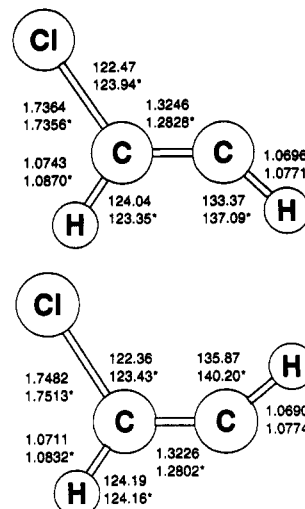


Figure 1. HF/6-31G* and MP2/6-31G* structures of cis- and trans- C_2H_2Cl . The MP2 structure is indicated by asterisks. Distances are in angstroms, and angles, in degrees.

TABLE I: Ab Initio Vibrational Frequencies and Principal Moments of Inertia for $C_2H_2Cl^{a,b}$

<i>trans</i> - C_2H_2Cl	<i>cis</i> - C_2H_2Cl	$C_2H_2^c$
Vibrational Frequencies (cm^{-1}) ^a		
<u>in plane</u>	<u>in plane</u>	<u>collinear</u>
3466 (3095)	3456 (3086)	3720 (3322)
3423 (3057)	3372 (3011)	3608 (3222)
1506 (1345)	1527 (1363)	2247 (2006)
1349 (1205)	1342 (1198)	
983 (878)	995 (888)	
762 (681)	740 (661)	
401 (358)	427 (381)	
<u>out of plane</u>	<u>out of plane</u>	<u>nonlinear</u>
898 (802)	845 (755)	882 (788)
673 (601)	653 (583)	793 (708)
Moments of Inertia ($amu \text{ \AA}^2$)		
85.9	87.5	13.9
77.5	81.0	13.9
8.42	6.51	

^a The vibrational frequencies were calculated at the HF/6-31G* level of theory. Anharmonic frequencies (in parentheses) were estimated by multiplying the harmonic frequencies by the 0.8929 scale factor.⁴⁹ ^b The moments of inertia were calculated at the MP2/6-31G* level of theory. ^c The experimental harmonic (anharmonic) frequencies for C_2H_2 are 3497 (3374), 3415 (3289), 2011 (1974), 747 (730), and 624 (612) cm^{-1} .⁴⁷

Pople and co-workers proposed a quantum chemical approach identified as the G1 method in which ab initio calculations at different levels of theory are combined in a systematic manner to account for full electron correlation and a complete basis set.⁴⁹ In very recent work⁵⁰ improvements to the G1 method were made to formulate the G2 method and this latter method was used to study the energetics of reaction 22. The trans configuration of C_2H_2Cl was found to have a zero point energy level 1.06 kcal/mol lower in energy than that of the cis configuration. Starting from the zero point level of *trans*- C_2H_2Cl , the G2 method gives $\Delta H^\circ_0 = 16.855$ kcal/mol for reaction 22, which is larger than the two values estimated above. As shown in the following sections, this larger G2 value for ΔH°_0 gives the best agreement between theory and experiment and was, thus, used for the calculations reported here.

B. Potential Energy Surface. Two different model potential energy surfaces were investigated for reaction 22 and are identified as models I and II. For each of the models the moments of inertia and anharmonic vibrational frequencies for C_2H_2Cl were assumed to be those in Table I for the trans configuration, which is the C_2H_2Cl configuration lower in energy.

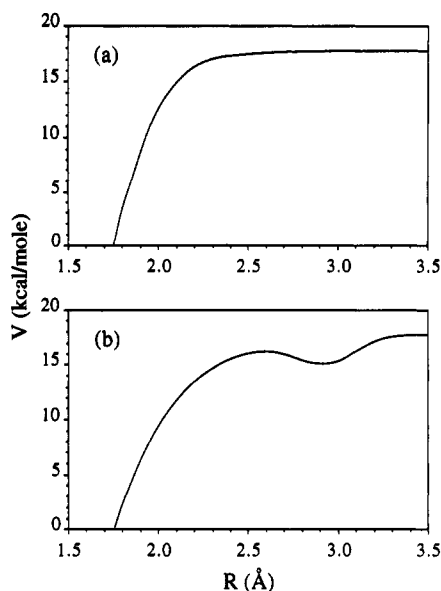


Figure 2. Classical potential energy curves for model I (a) and model II (b). R is the Cl - - C distance.

In considering the form of the reaction path potential energy curve for reaction 22, reference was made to previous *ab initio* calculations for the $C_2H_4Cl \rightarrow C_2H_4 + Cl$ reaction at HF/6-31G* and MP2/6-31G* levels of theory.⁵¹ Both of the calculations indicate the presence of a Cl - - C₂H₄ van der Waals minimum on the potential energy curve. However, given the incomplete basis sets and treatment of electron correlation for the calculations it cannot be unambiguously determined whether the actual potential has a van der Waals minimum and, thus, a saddlepoint. Also, if there is a saddlepoint, it cannot be deduced whether its potential energy is higher or lower than that of Cl + C₂H₄.

The model potentials I and II for $C_2H_2Cl \rightarrow Cl + C_2H_2$ were chosen with the above results for the $C_2H_4Cl \rightarrow Cl + C_2H_4$ reaction in mind. For model I the potential energy curve as a function of the forming C - - Cl bond distance is purely attractive with no Cl - - C₂H₂ van der Waals minimum and no saddlepoint. In contrast, model II has a van der Waals minimum and a saddlepoint. Qualitative potential energy curves for models I and II are shown in Figure 2. As described in the following, model I gives rise to a tight variational transition state for C₂H₂-Cl dissociation, while model II has a tight fixed transition state.

1. Model I: Variational Transition State. For model I the long-range potential between Cl and C₂H₂ is written as

$$V = V_{\text{intra}} + V_{\text{inter}} \quad (23)$$

where V_{intra} is the C₂H₂ intramolecular potential and V_{inter} is the Cl + C₂H₂ intermolecular potential.

The C₂H₂ intramolecular potential is written in terms of internal coordinates with the bends and stretches represented by harmonic oscillators

$$V_{\text{intra}} = \frac{1}{2}f_{rr}(r-r_0)^2 + \frac{1}{2}f_{rr}(\bar{r}-r_0)^2 + \frac{1}{2}f_{RR}(R-R_0)^2 + f_{rr}(r-r_0)(\bar{r}-r_0) + f_{rR}(r-r_0)(R-R_0) + f_{rR}(\bar{r}-r_0)(R-R_0) + \frac{1}{2}f_{\theta\theta}(\theta-\theta_0)^2 + \frac{1}{2}f_{\theta\theta}(\bar{\theta}-\theta_0)^2 \quad (24)$$

The coordinates r , R , and $\bar{r}$ are the H₁-C₁, C₁-C₂, C₂-H₂ distances, respectively, and θ and $\bar{\theta}$ are the H₁-C₁-C₂ and C₁-C₂-H₂ angles, respectively. Since an *ab initio* structure and *ab initio* frequencies are used for *trans*-C₂H₂Cl, for consistency the force constants and equilibrium coordinates for V_{intra} are taken from the *ab initio* calculations for C₂H₂. The MP2/6-31G* coordinates are $r_0 = 1.066$ Å, $R_0 = 1.216$ Å, and $\theta_0 = 180^\circ$. As discussed below, in

the calculations reported here anharmonic frequencies are required. Thus, anharmonic *ab initio* force constants were derived by multiplying the HF/6-31G* force constants by the scale factor (0.8929)². The resulting anharmonic force constants are $f_{rr} = 5.690$ mdyn/Å, $f_{RR} = 16.371$ mdyn/Å, $f_{rR} = -0.056$ mdyn/Å, $f_{r\bar{r}} = 0.005$ mdyn/Å, $f_{\theta\theta} = 0.291$ (mdyn Å)/rad², and $f_{\theta\bar{\theta}} = 0.089$ (mdyn Å)/rad².

To simplify the reaction path following calculations presented below, the nondiagonal bend-bend interaction term in eq 24 was neglected and the anharmonic $f_{\theta\theta}$ force constant was adjusted to 0.275 (mdyn Å)/rad² to obtain better agreement with the *ab initio* anharmonic frequencies. The C₂H₂ frequencies with $f_{\theta\theta} = 0.275$ (mdyn Å)/rad² and $f_{\theta\bar{\theta}} = 0.0$ are 3322, 3222, 2006, 827, and 671 cm⁻¹.⁵²

The Cl + C₂H₂ intermolecular potential, V_{inter} in eq 23, is represented by a Morse function for the Cl - - C forming bond and harmonic oscillators for the Cl - - C=C and Cl - - C-H bending motions; i.e.

$$V_{\text{inter}} = D_e[1 - e^{-\beta(r^a-r_0^a)}]^2 + \frac{1}{2}f_\phi(r^a)[\phi - \phi_0]^2 + \frac{1}{2}f_\chi(r^a)[\chi - \chi_0]^2 \quad (25)$$

Here r^a is the Cl - - C distance, ϕ the Cl - - C=C angle, and χ the Cl - - C-H angle. The equilibrium values for these coordinates are $r_0^a = 1.751$ Å and $\phi_0 = \chi_0 = 90^\circ$. The ϕ and χ bending force constants are written as a function of r^a so that they are attenuated as r^a is elongated; i.e.

$$f_\phi(r^a) = f_\phi^0 e^{-\alpha(r^a-r_0^a)^2} \quad r^a \geq r_0^a \\ = f_\phi^0 \quad r^a < r_0^a \quad (26)$$

The same expression is used for $f_\chi(r^a)$ except f_ϕ^0 is replaced by f_χ^0 . The values used for f_ϕ^0 and f_χ^0 , 1.076 and 0.9436 (mdyn Å)/rad², respectively, are the scaled *ab initio* force constants for *trans*-C₂H₂Cl. In eqs 25 and 26 β and α are two parameters that are adjusted to fit the experimental rate constants.

Since model I does not have a fixed transition state, variational transition states^{53,54} were found for the potential given by eqs 23-26. This required using the general chemical dynamics computer program VENUS⁵⁵ to find the reaction path⁵⁶⁻⁵⁸ for Cl + C₂H₂ → C₂H₂Cl* association and to calculate the principal moments of inertia and vibrational frequencies as a function of the path. The analytic functions used in eqs 23-26 are standard options in VENUS. The Cl - - C₂H₂ potential energy, principal moments of inertia, and vibrational frequencies as a function of the reaction path were used to place microcanonical transition states at the minima in the sum of states along the reaction path and canonical transition states at the maxima in the free energy along the reaction path.^{53,54}

2. Model II: Fixed Transition State. For the fixed transition state model, illustrated by the potential energy curve in Figure 2b, it was assumed the C₂H₂ moiety has the same geometry and vibrational frequencies as separated Cl and C₂H₂,⁴⁷ the Cl - - C=C angle is 90°, and the Cl - - C distance at the transition state is the same as that for the Cl - - C₂H₄ transition state at the HF/6-31G* level of theory (i.e., 2.6 Å). By making these assumptions there are only two unknown properties for the fixed Cl - - C₂H₂ transition state; the quantum potential energy difference E_{or}^* between separated Cl + C₂H₂ and the transition state and the frequency ν_{tr}^* for the Cl - - C₂H₂ transitional bending mode at the transition state. These two parameters are adjusted to fit the experimental rate constants.

3. Harmonic versus Anharmonic Frequencies. For the calculations reported here, of the Cl + C₂H₂ → C₂H₂Cl association rate constant $k_a(T,P)$, *ab initio* anharmonic frequencies are used for C₂H₂Cl. Thus, for consistency, the same type of frequencies are used for the Cl - - C₂H₂ transition state and acetylene. For

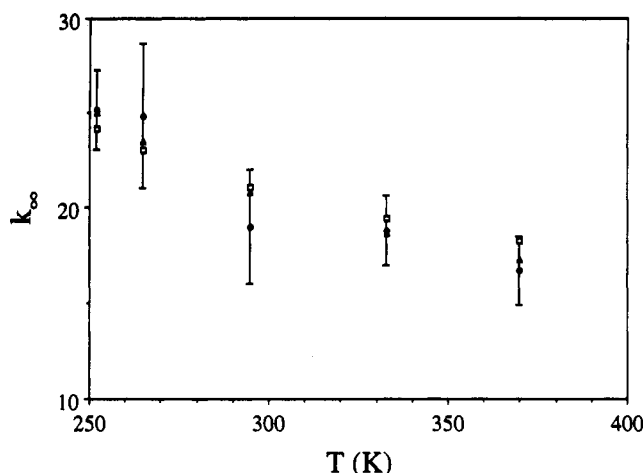


Figure 3. Plot of $k_a(T)$ in units of $10^{-11} \text{ cm}^3/(\text{molecule s})$ vs T (K): (O) (with error bars) experiment, eq 2a; (□) canonical calculations for model I, with $\beta = 7.0 \text{ \AA}^{-1}$ and $\alpha = 6.0 \text{ \AA}^{-2}$; (Δ) model II in Table III.

the calculations with model I, force constants for the C_2H_2 moiety in the transition state were chosen to match the ab initio anharmonic force constants for acetylene. For model II the frequencies for the C_2H_2 moiety in the transition state are the ab initio anharmonic frequencies for acetylene. It is worthwhile noting that when the high-pressure association rate constant $k_\infty(T)$ is calculated for model II, the type of frequency used for acetylene and the C_2H_2 moiety in the transition state does not matter, since the frequencies for these two species cancel in the calculations. This is not the case when $k_a(T)$ is calculated for Model I.

IV. Comparison of Experimental and Theoretical Association Rate Constants

The quantity measured experimentally is the rate constant for reaction 1 relative to the rate constant for the abstraction reaction $\text{Cl} + \text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_5 + \text{HCl}$.^{13,14} The recommended rate constant at 298 K from the NASA compilation⁴ for this abstraction reaction is $5.7 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$. For the temperature-dependent rate constant in the 220–350 K range, this compilation recommends the expression $7.7 \times 10^{-11} \exp(-90/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ with an uncertainty in E_a/R of ± 100 . In another analysis Kaiser¹⁴ found that the expression $8.68 \times 10^{-11} \exp(-113/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ fits all the abstraction rate constants for the 220–604 K temperature range to within $\pm 8\%$. It gives a rate constant of $5.94 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$ at 298 K. This latter temperature-dependent expression for the abstraction rate constant was used here to derive rate constants for reaction 1, i.e. $k_a(T, P)$, at all temperatures studied experimentally. The error in the ratio between $k_a(T, P)$ and the rate constant for $\text{Cl} + \text{C}_2\text{H}_6 \rightarrow \text{C}_2\text{H}_5 + \text{HCl}$ abstraction is quite small and represented by the scatter in $k_a(T, P)$ versus pressure for a particular temperature.^{13,14} In the following, RRKM theory is used to calculate values for $k_a(T, P)$ to compare with experiment.

A. Model I: Variational Transition State. For model I the transition-state structures used to calculate $k_a(T, P)$ are not fixed and, thus, must be chosen using variational transition-state criteria.^{53,54} In a microcanonical variational RRKM calculation of $k_a(T, P)$ transition states are found as a function of E and J if K is active, and E , J , and K if K adiabatic. A less detailed calculation involves a fixed temperature canonical variational calculation in which $k_\infty(T)$ is fit by a canonical transition state and then this transition state is used for all ensuing calculations of $k_a(T, P)$. As discussed in the Introduction, the Troe formalism was used to extrapolate measured rate constants for reaction 1 in the falloff region to the high-pressure limit to give $k_\infty(T)$ in eq 2a.¹³ This expression for $k_\infty(T)$ was fit with canonical transition states by varying the parameters β and α for model I, eqs 23–26. As shown in Figure 3, an excellent fit was obtained with $\beta = 7.0$

TABLE II: Comparison of $k_a(T)$ and $k_0(T)$ Rate Constants for Model I^a

T (K)	$k_a(T)^b$			$k_0(T)^b$		
	RRKM, canonical ^c	RRKM, micro-canonical ^d	RRKM, $E_{\min}^* \neq 0^e$	RRKM, canonical	RRKM, micro-canonical	RRKM, $E_{\min}^* \neq 0$
252	2.41	1.82	1.93	14.5	9.49	9.31
265	2.32	1.78	1.90	13.4	8.86	8.83
295	2.12	1.71	1.81	11.1	7.97	7.82
333	1.95	1.64	1.73	9.07	6.79	6.70
370	1.83	1.59	1.67	7.69	5.91	5.95

^a The rate constants were calculated using the angular momentum model with K active in both $\text{C}_2\text{H}_2\text{Cl}$ and the transition state, and for the potential energy surface with $\beta = 7.0 \text{ \AA}^{-1}$ and $\alpha = 6.0 \text{ \AA}^{-2}$. ^b $k_a(T)$ is in units of $10^{-10} \text{ cm}^3/(\text{molecule s})$ and $k_0(T)$ is in units of $10^{-30} \text{ cm}^6/(\text{molecule}^2 \text{ s})$. ^c Equation 13 is solved using a temperature-dependent canonical transition state. ^d Equation 13 is solved by finding a microcanonical transition state for each E, J pair. ^e Equation 13 is solved using a temperature-dependent canonical transition state, but the lowest transition-state energy is $E_{\min}^* = \Delta H^\circ_0 - E_0$, instead of zero.

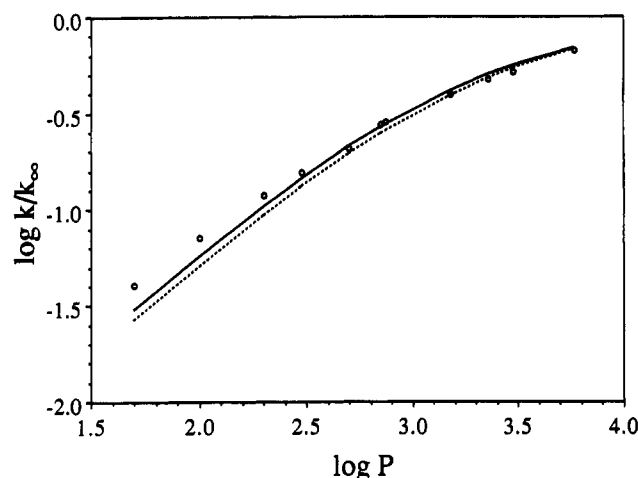


Figure 4. Plot of $\log [k_a(T, P)/k_a(T)]$ versus $\log P$ in Torr: (O) experiment; (—) canonical RRKM calculations using eq 13; (---) microcanonical RRKM calculation using eq 13. The temperature is 295 K.

$\text{\AA}^{-1}$ and $\alpha = 6.0 \text{ \AA}^{-2}$. The canonical transition state obtained in the fit is temperature dependent. At the median temperature, 295 K, the transition state has a Cl—C distance of 2.3 Å, a transitional mode bending frequency $\nu_{\text{tr}}^* = 196 \text{ cm}^{-1}$, and a potential energy 0.788 kcal/mol less than that of $\text{Cl} + \text{C}_2\text{H}_2$ (zero point energies are included in calculating this potential energy difference). The reaction path potential energy curve found with the fitted β and α values is plotted in Figure 2a as a function of the Cl—C bond length.

Microcanonical transition states were determined for the above model I potential energy surface (i.e., $\beta = 7.0 \text{ \AA}^{-1}$ and $\alpha = 6.0 \text{ \AA}^{-2}$) and used to calculate $k_a(T, P)$. Values of $k_\infty(T)$ and $k_0(T)$ calculated with the microcanonical transition states are compared in Table II with $k_\infty(T)$ and $k_0(T)$ determined with the canonical transition states. [The canonical $k_\infty(T)$ values are the same as those plotted in Figure 3.] The angular momentum model with K active in both $\text{C}_2\text{H}_2\text{Cl}$ and the transition state was used in these calculations. As found in previous work,²² the canonical $k_\infty(T)$ are slightly larger than the microcanonical $k_\infty(T)$, i.e. a factor 1.32 at 252 K and 1.15 at 370 K. However, what is particularly striking about the results in Table II is that the $k_\infty(T)/k_0(T)$ ratios agree to within 15% for the canonical and microcanonical RRKM calculations, which suggests that these two RRKM calculations should give similar $k_a(T, P)/k_\infty(T)$ curves. This was found to be indeed the case. The canonical and microcanonical $k_a(T, P)/k_\infty(T)$ curves calculated at 295 K from eq 13 are plotted in Figure 4, where they are compared with experiment. Approximate agreement with experiment is obtained by using $\beta_c = 0.37$ in both the canonical and microcanonical calculations. The

TABLE III: Parameters for the C_2H_2 - -Cl Tight Fixed Transition State (Model II)*

param	value	param	value
$r^*(\text{Cl}-\text{C})$ (Å)	2.6	ν_{tr}^* (cm^{-1})	325
$\phi^*(\text{Cl}-\text{C}=\text{C})$ (deg)	90	I_a^*, I_b^*, I_c^* (amu Å ²)	121.0, 107.7, 13.39
$\chi^*(\text{Cl}-\text{C}-\text{H})$ (deg)	90	E_{or}^* (kcal/mol)	-0.83

* The structure of C_2H_2 in the C_2H_2 - -Cl transition state is the same as that for the C_2H_2 molecule. The vibrational frequency for the transitional bending mode is given by ν_{tr}^* . E_{or}^* is the potential energy of the transition state minus the potential energy of $\text{Cl} + \text{C}_2\text{H}_2$. Zero point energies are included in calculating E_{or}^* .

excellent agreement found between the microcanonical and canonical $k_a(T,P)/k_\infty(T)$ curves indicates that the simpler canonical approach is sufficient for calculating the $\text{Cl} + \text{C}_2\text{H}_2$ association rate constant.

The variational transition state for model I has a tight structure similar to that for the fixed transition state of model II. Thus, as shown below, the two models give very similar fits to the experimental $k_a(T,P)$ rate constants. Extensive calculations were performed with the fixed transition-state model to compare the different models for treating the K quantum number and determine which gives the best fit to the experimental association rate constants.

For both the above canonical and microcanonical calculations of $k_a(T,P)$ the integration in eq 13 starts with zero energy in the transition state; i.e. $E_{\text{min}}^* = 0$. In the microcanonical calculations the unimolecular threshold E_0 correctly approaches the 0 K heat of reaction ΔH°_0 as $E_{\text{min}}^* \rightarrow 0$. However, for the canonical calculation E_0 is fixed at a value smaller than ΔH°_0 . Thus, as E_{min}^* approaches zero for the canonical calculation, the total energy of $\text{C}_2\text{H}_2\text{Cl}$ becomes less than ΔH°_0 . Thus, in the canonical calculation, states with insufficient energy to react are counted as reactive. This is a well-understood effect^{53,54,59} and explains why the canonical $k_a(T,P)$ are larger than the microcanonical values. To rectify this difference, one might be inclined to modify the canonical calculation so that the integration in eq 13 starts at $E_{\text{min}}^* = \Delta H^\circ_0 - E_0$, where E_0 is a function of temperature. Values of $k_\infty(T)$ and $k_0(T)$ calculated with such a model are listed in Table II. The $k_\infty(T)$ values are only 5–6% larger than those found from the microcanonical calculation, and thus, excellent agreement is found between the microcanonical and modified canonical $k_\infty(T)$ rate constants. This agreement is also maintained for the low-pressure limit, where the modified canonical rates are 1–2% smaller. Thus, the modified canonical calculation could be used to determine a $k_a(T,P)/k_\infty(T)$ curve which agrees with that determined by the microcanonical calculation.

B. Model II: Fixed Transition State. 1. Fitting from the High-Pressure Limit. To fit the high-pressure limit deduced from experiment, eq 2a, the transitional bending mode vibrational frequency ν_{tr}^* and the potential energy of the fixed transition state were adjusted. The resulting quantum potential energy (zero point energies are included) of separated $\text{Cl} + \text{C}_2\text{H}_2$ is 0.83 kcal/mol higher than that of the $\text{Cl} - \text{C}_2\text{H}_2$ transition state, and ν_{tr}^* equals 325 cm^{-1} . These parameters along with others for the fixed transition state are summarized in Table III. The $k_\infty(T)$ curve calculated with this model is plotted in Figure 3, where it is compared with experiment, eq 2a, and the calculation with the canonical variational transition-state model. Particularly good agreement is seen between the variational and fixed transition-state calculations.

Using this fixed transition state, $k_a(T,P)$ was calculated for the models with K either active or adiabatic in both $\text{C}_2\text{H}_2\text{Cl}$ and the $\text{Cl} - \text{C}_2\text{H}_2$ transition state. The calculations were performed at the five temperatures studied experimentally; i.e. 252, 265, 295, 333, and 370 K.^{13,14} Values were chosen for the collision efficiency β_c to obtain agreement between the experimental and theoretical $k_a(T,P)$ curves. For the calculations with K active in

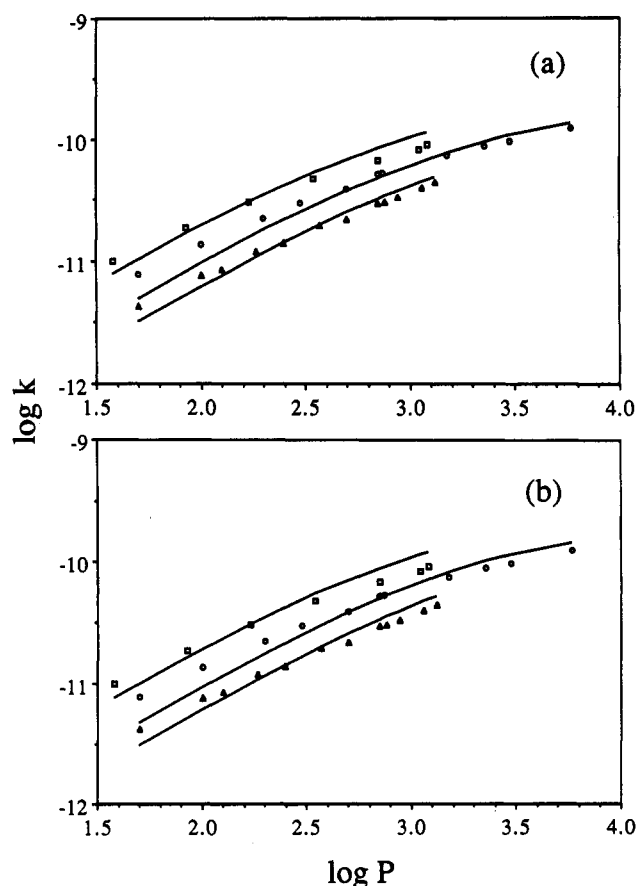


Figure 5. Plot of $\log [k_a(T,P)]$ versus $\log P$. k_a is in units of $\text{cm}^3/(\text{molecule s})$, and P is in units of Torr. Key: ($\square$) experiment at 252 K; ($\circ$) experiment at 295 K; (Δ) experiment at 370 K. The curves in (a) are calculations for model II with K active in both $\text{C}_2\text{H}_2\text{Cl}$ and the transition state. In (b) K is adiabatic for both of these species.

both the transition state and $\text{C}_2\text{H}_2\text{Cl}$, the fitted values for β_c are 0.30, 0.29, 0.23, 0.33, and 0.28 for $T = 252, 265, 295, 333$, and 370 K, respectively. However, treating K as adiabatic requires higher β_c values of 0.52, 0.50, 0.38, 0.52, and 0.44 for the same respective temperatures as above. Comparisons between the theoretical and experimental $k_a(T,P)$ values at 252, 295, and 370 K are given in Figure 5a for the calculations with the K active model and in Figure 5b for the calculations with the K adiabatic model. Overall, there is fairly good agreement between the experimental and calculated $k_a(T,P)$.

A significant finding from these calculations is that the value of β_c required to fit an experimental $k_a(T,P)$ curve depends on how the K quantum number is treated in the RRKM calculations. Making K adiabatic for both the transition state and unimolecular reactant requires increasing β_c by ~ 60 –70%. In the next section calculations are presented which consider all four models for treating the K quantum number (eqs 4 and 6–8).

2. Fitting from the Low-Pressure Limit. The low-pressure rate constant deduced from experiment, eq 2b, can be fit with the RRKM expressions discussed in section II.B by adjusting the collision efficiency β_c . The two models with K active in the transition state give $k_0(T)$ in eq 19. With K adiabatic in the transition state the $k_0(T)$ expression depends on how K is treated in $\text{C}_2\text{H}_2\text{Cl}$. If K is adiabatic for $\text{C}_2\text{H}_2\text{Cl}$, $k_0(T)$ is given by eq 20. However, for K active in $\text{C}_2\text{H}_2\text{Cl}$, $k_0(T)$ is given by eq 21. Listed in Table IV are the values of β_c obtained when $k_0(T)$ in eq 2b is fit with eqs 19–21. It is striking that the different models for treating K require different $\beta_c(T)$ to fit $k_0(T)$. As found above in fitting from the high-pressure limit, the $\beta_c(T)$ required to fit $k_0(T)$ when K is adiabatic in both the transition state and $\text{C}_2\text{H}_2\text{Cl}$ is approximately 70–80% higher than when K is active in both these species. However, the largest values of β_c are for

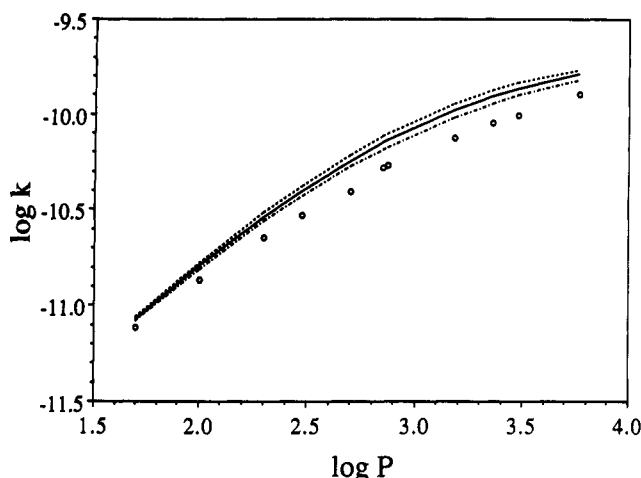


Figure 6. Plot of $\log k_a$ versus $\log P$ at 295 K. The units for k_a and P are the same as in Figure 5. $\circ$ are the experimental points. The calculated curves are for model II with β_c chosen to fit $k_0(T)$ in eq 2b and the transition state chosen to fit $k_\infty(T)$ in eq 2a. Key: (—) K active/active model; (---) K adiabatic/adiabatic model; (- - -) K active/adiabatic model.

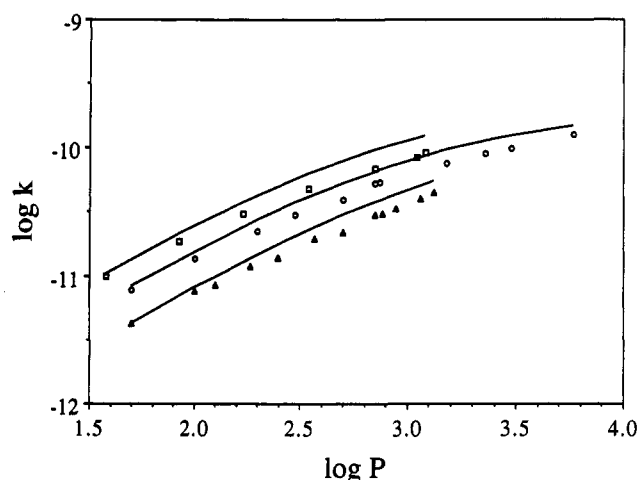


Figure 7. Same plots as in Figure 6, but only for the K active/adiabatic model; ($\square$) 252 K; ($\circ$) 295 K; (Δ) 370 K.

the model with K adiabatic in the transition state and active in the molecule. They are twice the size of the β_c needed when K is active in the transition state.

Using the values for β_c in Table IV and the tight fixed transition-state model discussed above, which fits $k_\infty(T)$ in eq 2a, the four different treatments of the K quantum number were used to calculate $k_a(T,P)$. The rate constants calculated at 295 K are plotted in Figure 6, where they are compared with the experimental rate constants. The expressions with K adiabatic in the transition state, eqs 14 and 15, give nearly the same $k_a(T,P)$ for the pressures studied experimentally, and the rate constants for the adiabatic/active model are not included in this figure. Neither one of the four models for treating K gives good agreement with experiment. However, of the four models the one with K active in the transition state and adiabatic for the molecule, eq 16, gives the best agreement with experiment. This model was further tested by using it to calculate values of $k_a(T,P)$ for all pressures and temperatures studied experimentally,^{13,14} and a slightly improved agreement with experiment was found at the higher temperatures of 333 and 370 K. The $k_a(T,P)$ curves calculated at $T = 252$, 295, and 370 K for the active/adiabatic model are shown in Figure 7.

An attempt was made to obtain a better fit to the experimental $k_a(T,P)$, with the active/adiabatic model, by maintaining the fit to the extrapolated $k_0(T)$ in eq 2b, but not fitting the extrapolated $k_\infty(T)$ in eq 2a and varying β_c and the transition-state parameters

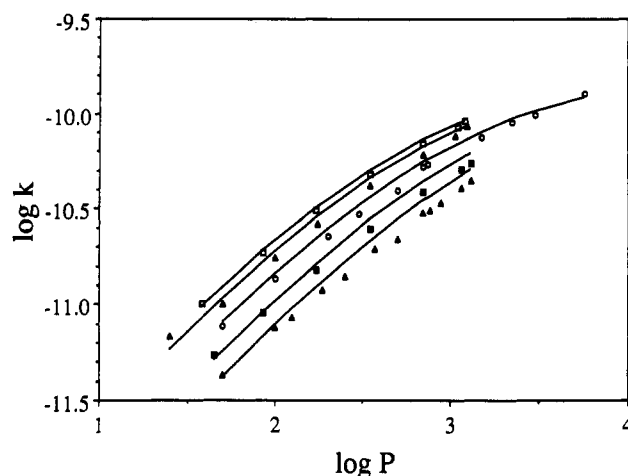


Figure 8. Same plots as in Figure 7 for the K active/adiabatic model, but the transition structure and β_c have been modified to obtain better agreement with the experimental $k_a(T,P)$ (see text): ($\square$) 252 K; (Δ) 265 K; ($\circ$) 295 K; ($\blacksquare$) 333 K; ($\blacktriangle$) 370 K.

E_{or}^* and ν_{tr}^* . The type of fit obtained, for a transition state with $E_{or}^* = -0.39$ kcal/mol and $\nu_{tr}^* = 125$ cm^{-1} , is illustrated in Figure 8. (The β_c values for the calculated curves are 0.88, 0.84, 0.78, 0.70, and 0.63 for T of 252, 265, 295, 330, and 370 K, respectively). Improved agreement with experiment is clearly found, particularly at the lower temperatures. A similar result is found for the other models for treating K , if the transition-state structure and β_c are varied.

It is worthwhile reviewing the above analysis. First, to fit $k_\infty(T)$ in eq 2b either a tight variational or tight fixed transition state is required, which has a potential energy less than that of the $\text{Cl} + \text{C}_2\text{H}_2$ reactants. Second, by using such a transition state and not attempting to fit $k_0(T)$ in eq 2b, the experimental $k_a(T,P)$ can be approximately reproduced with $\beta_c \sim 0.29$ for the active/active model and $\beta_c \sim 0.47$ for the adiabatic/adiabatic model. Third, when $k_0(T)$ in eq 2b is fitted using all four models for treating K , the resulting values for β_c vary from ~ 0.4 for the two models with K active in the transition state to ~ 1.0 for the adiabatic/active model. Fourth, in fitting the experimental $k_a(T,P)$ with the β_c values deduced by fitting $k_0(T)$ in eq 2b and the transition-state structure which fits $k_\infty(T)$ in eq 2a, poor agreement with experiment is found for all models for treating K . Fifth, by retaining the fit to $k_0(T)$, but varying the transition-state parameters, the experimental $k_a(T,P)$ curves are well fit at 252, 265, and 295 K, with poorer fits at the higher temperatures. However, with these variations in the transition-state properties, $k_\infty(T)$ in eq 2a, deduced from experiment, is no longer fit and β_c has values between 0.6 and 0.9.

V. Discussion

There are three major issues considered in this work: (1) the sensitivity of RRKM theory calculations to the treatment of the K quantum number, (2) the extent of agreement between RRKM and experimental rate constants for $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$ association as a function of pressure and temperature, and (3) the nature of the reaction path potential energy curve for $\text{Cl} + \text{C}_2\text{H}_2$ association. Each of these issues is discussed below.

A. Sensitivity of RRKM Calculations to the Treatment of the K Quantum Number. One of the most exciting (or discouraging, depending on one's point of view) results of this study is the sensitivity of the pressure- and temperature-dependent RRKM rate constant for $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$ association to the treatment of the K quantum number. Four models are considered here for treating K . In previous work K has been assumed to be either active or adiabatic for both the unimolecular reactant and transition state. In this paper, two additional models are considered by allowing K to be treated differently in the transition

state than in the reactant, e.g. active in the transition state but adiabatic in the molecule.

At the high-pressure limit each of the four models for treating K gives the same association rate constant, if the same transition-state structure is used for each. At the low-pressure limit, the two models with K active in the transition state give the same association rate constant. However, with K adiabatic in the transition state, the low-pressure limiting rate constant depends on whether K is treated as adiabatic or active for the molecule and differs from that with K active in the transition state. Thus, there are three different expressions for the low-pressure rate constant $k_0(T)$ each of which requires a different collision efficiency $\beta_c(T)$ to fit experiment. Using the $\beta_c(T)$ chosen to fit the experimental $k_0(T)$, the four models for treating the K quantum number are expected to give different rate constants at intermediate pressures.

It should be pointed out that so far in this article another model for treating K as an active degree of freedom in both the transition state and molecule has not been considered. This model was originally proposed by Rabinovitch and co-workers^{60,61} and has been advanced by Forst^{27b,35} and Gilbert and Smith.⁴⁰ In this model the energy for the K -dependent terms in eq 3, i.e.

$$E_r(K) = \frac{K^2 \hbar^2}{2} [I_c^{-1} - (I_c^{-1} + I_b^{-1})/2] \quad (27)$$

is assumed to be active and exchange freely with the vibrational degrees of freedom. The sum of states for the transition state is then written as a convolution between the sum of states for the internal degrees of freedom and the density of states for the active external rotation. The density of states for the molecule is written similarly, except a density of states is used for the internal degrees of freedom. The resulting expressions for the sum and density are

$$G(E^*) = \int_0^{E^*} G_v(E) N_r(E^* - E) dE \quad (28a)$$

$$N(E_v) = \int_0^{E_v} N_v(E) N_r(E_v - E) dE \quad (28b)$$

where E_v is the active energy for the energized reactant and E^* is the active energy of the transition state. N_r in eqs 28a and 28b is the classical density of states for a 1-dimensional rotor. The adiabatic rotational energies $E_r(J)$ and $E_r^*(J)$ are given by the J -dependent term in eq 3. Since the total energy is $E = E_v + E_r(J) = E^* + E_r^*(J) + E_0$, the unimolecular rate constant for this model can be written as

$$k(E, J) = \frac{1}{h} \frac{G[E - E_r^*(J) - E_0]}{N[E - E_r(J)]} \quad (29)$$

A shortcoming of this model is that the proper restrictions are not placed on K , i.e. $-J \leq K \leq J$. As a result of the free exchange of energy between vibrational modes and the K -dependent term, the angular momentum about the symmetry axis may exceed J . Coriolis coupling allows the $2J + 1$ K states to mix, but this model places no restrictions on this mixing.

Regardless of possible shortcomings of this model, defined by eqs 27–29, it is still of interest to apply it to reaction 22. For a fixed transition state it is straightforward to show (see the Appendix) that this model gives the same $k_\infty(T)$ as that in eq 18 and, thus, will fit the $k_\infty(T)$ expression in eq 2a with the transition-state structure in Table II. The pressure-dependent association rate constant is given by eq 13, for which $k(E, J)$ in eq 29 is used for $k_{-1}(E, J)$ and $G(E^*)$ in eq 28a is used for $G^*(E, J)$. The resulting expression for $k_a(T, P)$, along with the fixed transition-state structure in Table II and the G2 value for ΔH°_0 of 16.855 kcal/mol, can be used to fit the experimental pressure and the temperature-dependent association rate constants by varying β_c .

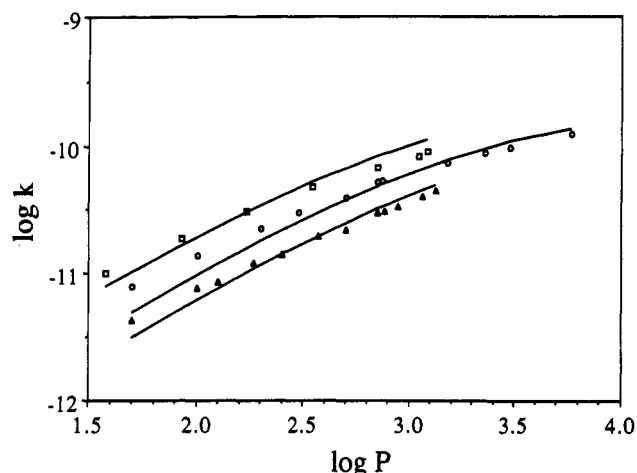


Figure 9. Similar to the calculations in Figure 5 for Model II, except K is treated active as defined by eqs 27–29.

TABLE IV: Values for β_c Which Fit $k_0(T)$ in Equation 2b

T (K)	treatment of K (transition state/molecule)			
	active/ active	adiabatic/ adiabatic	adiabatic/ active ^a	active/ adiabatic
252	0.41	0.75	0.99	0.41
265	0.41	0.74	0.98	0.41
295	0.41	0.71	0.95	0.41
333	0.40	0.68	0.92	0.40
370	0.39	0.64	0.88	0.39
$\beta_c(T)$	$0.852T^{-0.13}$	$8.22T^{-0.43}$	$5.01T^{-0.29}$	$0.852T^{-0.13}$

^a K is adiabatic for the transition state and active for the molecule $\text{C}_2\text{H}_2\text{Cl}$.

(T).⁶² The fitted values for β_c are 0.25, 0.25, 0.20, 0.28, and 0.25 for $T = 252, 265, 295, 333$, and 370 K, respectively. The fits at 252, 295, and 370 K are shown in Figure 9. To fit $k_0(T)$ in eq 2b requires β_c of 0.35, 0.35, 0.36, 0.36, and 0.35 for T of 252, 265, 295, 333, and 370 K, respectively. These values for β_c are only 10% smaller than those obtained above with the active/active model (e.g., see Table IV). It is not clear that such agreement between the two K active models is a general result. For other reactions they give considerably different values for $k(E, J)$.^{24,25}

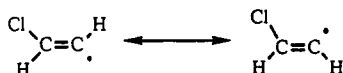
With the four models presented in section II and the model described above, there are at least five different models for treating the K quantum number in RRKM calculations. When a pressure-dependent association rate constant is being fit, each model may require a different collisional stabilization efficiency β_c . This result indicates that until a proper model for treating K is identified it will be difficult to determine unambiguous values for β_c by fitting experimental data.

B. Ability of RRKM Theory To Fit the Experimental $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$ Association Rate Constant. To calculate the low pressure $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$ association rate constant $k_0(T)$ requires knowing (1) the structure and vibrational frequencies of $\text{C}_2\text{H}_2\text{Cl}$, (2) ΔH°_0 for $\text{C}_2\text{H}_2\text{Cl}$ dissociation, and (3) the collisional stabilization efficiency β_c . In this work the structure and vibrational frequencies of $\text{C}_2\text{H}_2\text{Cl}$ are determined by MP2/6-31G* and HF/6-31G* ab initio calculations, respectively. An ab initio G2 calculation gives $\Delta H^\circ_0 = 16.855$ kcal/mol. By comparing G2 theory and experimental ΔH°_0 values for a large number of reactions, it has been estimated that G2 theory gives ΔH°_0 with a maximum uncertainty of 2.0 kcal/mol. To compare with the G2 calculation, two semiempirical estimates equal to 16.2 and 14.5 kcal/mol were made for ΔH°_0 of reaction 22 by considering heats for reactions similar to reaction 22; see section III.A. These estimates suggest that, if the G2 value for ΔH°_0 is in error, it is too high. Using the ab initio $\text{C}_2\text{H}_2\text{Cl}$ structure and energy, values for β_c were adjusted to fit the experimentally

derived $k_0(T)$ in eq 2b. The resulting β_c values vary from ~ 0.4 for the two models which treat K active in the transition state to ~ 1.0 for the model with K adiabatic in the transition state and active for the molecule. Previous estimates of β_c for the bath gas air considered here range from 0.1 to 0.5.^{27b,40,63}

From a recent study,⁶⁴ closely related to that reported here, it has been found that large values of β_c are required to interpret collisional stabilization probabilities of chlorinated hydrocarbons. Thus, this study and that reported here suggest that large values of β_c may be needed to model collisional stabilization of excited molecules containing chlorine atoms. Large values of β_c indicate that, on average, large amounts of energy are transferred from the excited molecule; i.e. $\langle \Delta E_{\text{down}} \rangle$ values. Using the formalism presented by Troe,⁶⁵ it is found that for $\text{C}_2\text{H}_2\text{Cl}$ deactivation at 300 K β_c values of 0.5, 0.6, and 0.7 correspond to $\langle \Delta E_{\text{down}} \rangle$ values of 589, 838, and 1250 cm^{-1} , respectively.

As shown by the equations in section III.B, k_0 at a particular temperature is proportional to both β_c and the Boltzmann weighted density of states for $\text{C}_2\text{H}_2\text{Cl}$. Thus, if the density of states were increased, a smaller value of β_c could be used to fit k_0 . To increase the density of states requires increasing ΔH°_0 for reaction 22 and/or including additional anharmonicity for $\text{C}_2\text{H}_2\text{Cl}$. The above discussion indicates that the G2 value for ΔH°_0 used here, if in error, is too large. Nevertheless, increasing ΔH°_0 by its suggested maximum uncertainty of 2 kcal/mol would lower the derived β_c values by $\sim 40\%$. An anharmonic density of states is calculated here for $\text{C}_2\text{H}_2\text{Cl}$ by using anharmonic $n = 0 \rightarrow 1$ transition frequencies. However, the $\text{C}_2\text{H}_2\text{Cl}$ molecule has other anharmonic properties which may increase its density of states. One is the double-well potential for H-atom inversion; i.e.



When the energy in the $\text{C}=\text{C}-\text{H}$ bend mode is less than the inversion barrier, each vibrational level will be split into two sublevels. This will increase the density of states 2-fold. For energies in the $\text{C}=\text{C}-\text{H}$ bend in excess of the inversion barrier, the effect of the inversion double well is less clear. However, at these energies the spacings between the $\text{C}=\text{C}-\text{H}$ bend levels are expected to be less than the harmonic spacings for one of the potential minima, so that the $\text{C}_2\text{H}_2\text{Cl}$ density of states will be larger than that of the harmonic model. Thus, for both low and high energy in the $\text{C}=\text{C}-\text{H}$ bend, the effect of the double-well potential for H-atom inversion is to increase the $\text{C}_2\text{H}_2\text{Cl}$ density of states. Detailed calculations will be required to determine the quantitative nature of this anharmonicity effect.^{66,67}

Recent studies have shown that anharmonic corrections can be very important. For both H_2CO and C_2H_2 at ~ 80 kcal/mol the experimental density of states is 10–12 times larger than the harmonic estimate.^{32,68} Including first-order anharmonicities with a Dunham-type expansion only increases the calculated density by a factor of 2. Apparently, both H_2CO and C_2H_2 have higher-order anharmonicities^{69,70} which become important at high energies. A recent theoretical study of HCN found that it has an anharmonic density of states at its dissociation energy 8-fold larger than the harmonic density.⁷¹

The above discussions suggest that it may be possible to fit $k_0(T)$ in eq 2b deduced from experiment with reasonable values for β_c . To extend the fit of the $\text{Cl} + \text{C}_2\text{H}_2$ association rate constant to the intermediate pressures studied experimentally requires knowing the $\text{Cl}-\text{C}_2\text{H}_2$ transition-state structure and vibrational frequencies in addition to ΔH°_0 , β_c , and the structure and vibrational frequencies for $\text{C}_2\text{H}_2\text{Cl}$. Two approaches were used to extend the above fit from the low-pressure limit to intermediate pressures. In one approach the transition state that fits $k_\infty(T)$ in eq 2a was used in the calculations. None of the models for treating K gave acceptable agreement with experiment. The

model with K active in the transition state and adiabatic in $\text{C}_2\text{H}_2\text{Cl}$ gave the best agreement with experiment. However, even with this model, only at 333 and 370 K was there a semblance of agreement between theory and experiment. In the second approach β_c and the transition-state parameters $E_{0^\ddagger}$ and $\nu_{\text{tr}}^\ddagger$ were arbitrarily varied to obtain a “best” fit to experiment. As shown in Figure 8, an improved agreement with experiment was found, particularly for $T \leq 295$ K. However, the effects of this fit are β_c values which range between 0.6 and 0.9 and transition-state parameters which do not reproduce $k_\infty(T)$ in eq 2a. In previous RRKM calculations for other reactions,^{40,61,72–76} β_c values obtained from fitting $k_0(T)$ have been used to reproduce $k_a(T,P)$ curves at intermediate pressures. For the $\text{Cl} + \text{C}_2\text{H}_2 \rightleftharpoons \text{C}_2\text{H}_2\text{Cl}$ system this procedure appears not to work.

Another attempt was made to reproduce the $\text{Cl} + \text{C}_2\text{H}_2$ association rate constant at intermediate pressures by fitting from the high-pressure limit. A transition state was chosen (Table II) which reproduces $k_\infty(T)$ in eq 2a, and then $\beta_c(T)$ was varied to fit the experimental rate constants. The resulting best fit values for β_c are approximately 50% smaller than those which fit $k_0(T)$. Also, the fits to the experimental $k_a(T,P)$ curves are only approximate since the curvatures of the curves are not reproduced.

The approaches used here for fitting the pressure and temperature dependence of reaction 22 are very similar to those used in many previous studies.^{40,61,72–76} However, it is uncertain that complete agreement between RRKM theory and experiment can be attained. It is possible that, when an energy-dependent anharmonic density of states is used for $\text{C}_2\text{H}_2\text{Cl}$ and the transition state and RRKM theory is combined with a master equation formalism⁴⁰ to calculate the pressure-dependent association rate constant, agreement with experiment will be obtained.

Complete agreement with the experimental $k_a(T,P)$ at intermediate pressures could be obtained with the formalism used here by not attempting to fit the high- and low-pressure limiting rate constants in eqs 2a and 2b. These expressions for $k_\infty(T)$ and $k_0(T)$ were derived^{13,14} by using the Troe shape factor F_{cent} .¹⁵ A value of $F_{\text{cent}} = 0.6$ was used to fit the data at 295 K.¹³ This is the room-temperature value of F_{cent} recommended for reaction 1 and similar reactions in a recent NASA compilation.⁴ Thus, the inability to simultaneously fit k_∞ , k_0 , and the rate constants at intermediate pressures for $T = 295$ K is somewhat troublesome. The same F_{cent} value was used to determine k_∞ and k_0 for the other temperatures studied experimentally,¹⁴ i.e. 252, 265, 333, and 370 K. Though F_{cent} is a weak function of temperature,^{77–79} its actual temperature dependence for reaction 1 is unknown.¹⁴ Thus, less than complete agreement might be expected with the k_∞ and k_0 rate constants deduced for temperatures other than 295 K.

Given previous studies, it might not be unusual to find that RRKM theory does not represent the $\text{Cl} + \text{C}_2\text{H}_2$ association dynamics. As discussed in the Introduction, from measurements of product energy partitioning it has been proposed that the RRKM assumption of complete intramolecular vibrational energy redistribution is not valid for reaction systems very similar to the $\text{Cl} + \text{C}_2\text{H}_2$ system.^{17–20} Also in a detailed study, akin to that reported here, agreement between RRKM theory and experiment is not found for $\text{H} + \text{C}_2\text{H}_2 \rightleftharpoons \text{C}_2\text{H}_3$.⁸⁰ Finally, RRKM theory does not seem to reproduce the experimental kinetics for the XO ($\text{X} = \text{Cl}, \text{Br}, \text{I}$) + NO_2 reactions.^{81,82}

It is possible that to accurately model the kinetics of $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$ one will have to consider a reaction occurring via the preparation, decomposition, and collisional stabilization of $\text{C}_2\text{H}_2\text{Cl}$ compound state resonances.^{83–85} Such behavior is not inconceivable since the density of states of $\text{C}_2\text{H}_2\text{Cl}$ is quite low; e.g. at the unimolecular threshold with $J = 0$ the state density is 4 states/ cm^{-1} . There are two limiting patterns for the unimolecular rate constants of compound state resonances.⁸⁴ There may be random fluctuations in the rate constants, a pattern

consistent with chaotic intramolecular dynamics. In the other limiting case there is mode-specific behavior in which quantum numbers can be assigned to the wave function for the resonance states and, thus, the lifetimes of the resonance states (e.g. short or long) can be associated with mode-specific behavior. In future work it will be of interest to determine if either of these two limiting cases is applicable to $\text{C}_2\text{H}_2\text{Cl}$.

C. $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$ Potential Energy Curve. The experimental negative temperature dependence of the $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$ association rate constant^{13,14} indicates that $\text{Cl} + \text{C}_2\text{H}_2$ has an attractive potential and that the transition state has a potential energy less than that of $\text{Cl} + \text{C}_2\text{H}_2$. One can envision two different types of potential energy curves for $\text{Cl} + \text{C}_2\text{H}_2$ association. In one the potential is purely attractive without a maximum. For the second there is a long-range van der Waals minimum and then a saddlepoint (maximum) with a potential energy less than that of $\text{Cl} + \text{C}_2\text{H}_2$. The calculations reported here do not differentiate between these two types of potential energy curves, since each can accommodate the tight transition state required to fit the association rate constants. It is of interest to note that the existence of a van der Waals minimum along the reaction path is consistent with the structure of the $\text{Ar} \cdots \text{C}_2\text{H}_2$ van der Waals complex.⁸⁶ However, determining the actual form of the $\text{Cl} + \text{C}_2\text{H}_2 \rightarrow \text{C}_2\text{H}_2\text{Cl}$ potential energy curve will require extensive multireference ab initio calculation, which will be an important component of future studies.

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Appendix

The high-pressure rate constant for the Gilbert-Smith model is

$$k_\infty(T) = \frac{\sigma}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE \sum_{J=0}^{\infty} (2J+1) G(E^* - E_r^*) e^{-E/k_B T} \quad (\text{A1})$$

where $G(E^*)$ is given by eq 28a and E_r^* is the $E_r^*(J)$ of eq 29. If the transition state is the same for all E and J , $E = E^* + E_0$ and

$$k_\infty(T) = \frac{\sigma e^{-E_0/k_B T}}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE^* \sum_{J=0}^{\infty} (2J+1) G(E^* - E_r^*) e^{-E^*/k_B T} \quad (\text{A2})$$

When the spacings between the energy levels are small compared to $k_B T$, the vibrational/rotational energy levels can be treated as continuous, so eq A2 becomes

$$k_\infty(T) = \frac{\sigma e^{-E_0/k_B T}}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE^* \int (2J+1) G(E^* - E_r^*) e^{-E^*/k_B T} dJ \quad (\text{A3a})$$

$$= \frac{\sigma e^{-E_0/k_B T}}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int dE^* \int (2J+1) e^{-E_r^*/k_B T} dJ \times \int G(E_a^*) e^{-E_a^*/k_B T} \Delta(E^* - E_r^* - E_a^*) dE_a^* \quad (\text{A3b})$$

$$= \frac{\sigma e^{-E_0/k_B T}}{hQ_{\text{C}_2\text{H}_2}Q_{\text{rel}}} \int (2J+1) e^{-E_r^*/k_B T} dJ \times \int G(E_a^*) e^{-E_a^*/k_B T} dE_a^* \quad (\text{A3c})$$

where $E^* = E_r^* + E_a^*$ is the energy for the active degrees of freedom, which is vibrational motion and 1-dimensional external rotation motion.

The first integral in eq A3c is the partition function for 2-dimensional external rotation

$$Q_{\text{rot},2D}^* = \int (2J+1) e^{-E_r^*/k_B T} dJ = 8\pi^2 k_B T (I_a I_b)^{1/2} / h^2 \quad (\text{A4})$$

where $I_a = I_b \neq I_c$. The second integral in eq A3c can be separated into two parts, as is done in eq A3. These parts are for vibrational motion and 1-dimensional external rotational motion and are written as

$$\int N(E_{1D}^*) e^{-E_{1D}^*/k_B T} dE_{1D}^* \int G(E_v^*) e^{-E_v^*/k_B T} dE_v^* \quad (\text{A5})$$

where $E_{1D}^* + E_v^* = E_a^*$, $N(E_{1D}^*)$ is the density of states for the 1-dimensional external rotation, and $G(E_v^*)$ is the sum of states for the vibrational modes. The first integral in eq A5 is the partition function for the 1-dimensional external rotation

$$Q_{\text{rot},1D}^* = (8\pi^2 I_c k_B T / h^2)^{1/2} \sqrt{\pi} \quad (\text{A6})$$

The second integral in eq A5 is the vibrational partition function for the transition state multiplied by $k_B T$; i.e. $Q_{\text{vib}}^* k_B T$. Thus, by combining this result with eqs A4–A6, eq A3c can be written as

$$k_\infty(T) = \frac{\sigma k_B T}{Q_{\text{C}_2\text{H}_2} Q_{\text{rel}}} Q_{\text{rot},2D}^* Q_{\text{rot},1D}^* Q_{\text{vib}}^* e^{-E_0/k_B T} = \frac{\sigma k_B T}{Q_{\text{C}_2\text{H}_2} Q_{\text{rel}}} Q_{\text{rot}}^* Q_{\text{vib}}^* e^{-E_0/k_B T}$$

This is the same expression for $k_\infty(T)$ as in eq 18, which was derived for the model with K treated explicitly with its proper constraints, i.e. $-J \leq K \leq J$.

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